

FRC 100.009: Local Reciprocity Dynamics

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Abstract

Develops a local field-level formulation of the Adaptive Resonance Framework (ARF) within Fractal Resonance Cognition (FRC). We formulate local reciprocity balance, storage, and flux, and define ARF as a state-dependent activation operator that gates transport based on local reciprocity defect, validating it through a numerical toy model.

1. Introduction

The global entropy--coherence reciprocity relation established in [[FRC-566-001]] is **directional**, not a closed conservation law. The secure content is that total entropy production is non-negative ($S_{\text{total}} \geq 0$), while the combined ledger $S_{\text{total}} + k^* d \ln C$ is

bounded below only when coherence is created and carries no floor when coherence is destroyed slowly (566.001 v2.0; measured in 100.005 v2.1). The closed form

$$\left[dS + k^* d \ln C = 0 \right]$$

is the reversible, locked-limit boundary case only, relating informational entropy S and coherence $C \in (0, 1]$. A key structural gap remains: how does this relation appear in a local, open, distributed dynamical system with local storage, flux, source, sink, and defect structure? The local reciprocity defect δ_R developed below is precisely the local reciprocity *current* — expected nonzero exactly where the system is open and driven, and vanishing only in the closed, silent equilibrium.

This paper develops a local field-level formulation of the **Adaptive Resonance Framework (ARF)**. We localize the global reciprocity law into a field continuity-style balance equation and treat the activation operator as derived from local reciprocity defect rather than inserted as an arbitrary constant floor.

2. Core Mathematical Formulation

Let $x \in \Omega \subset \mathbb{R}^d$ denote position and $t \geq 0$ time.

2.1 Storage and Flux

We define the **local reciprocity storage density** as:

$$\left[\Phi_R(x,t) = s(x,t) + k^* \ln \big(C(x,t) + \epsilon \big) \right]$$

where $s(x,t)$ is the local entropy-like density, $C(x,t)$ is the local coherence, and $\epsilon > 0$ is a small regularization parameter.

The **reciprocity flux** $J_R(x,t)$ is closed via a local gradient diffusion law:

$$\left[J_R = -D_R \nabla \Phi_R \right]$$

with diffusivity $D_R > 0$.

2.2 Local Reciprocity Defect

The open-system local **reciprocity defect** $\delta_R(x,t)$ measures the departure from local balance:

$$\left[\delta_R(x,t) = \partial_t \Phi_R + \nabla \cdot J_R - \Sigma_R(x,t) \right]$$

where $\Sigma_R(x,t)$ is the external reciprocity source density.

To avoid circular dependency in numerical settings, the time derivative of storage is computed using a lagged backward finite-difference discretization:

$$\frac{\partial_t \Phi_R^n \approx \frac{\Phi_R^n - \Phi_R^{n-1}}{\Delta t}}$$

2.3 ARF Gating Throttle

The Adaptive Resonance gating throttle $\mathcal{A}(x,t) \in [0, 1]$ acts as a state-dependent activation or mobility operator:

$$\mathcal{A}(x,t) = \tanh\big(\kappa R(x,t) C(x,t) |\delta_R(x,t)|\big)$$

where $\kappa > 0$ represents the defect sensitivity, and $R(x,t) = 1.0$ is the constant receptivity field. Transport and transformations only become active in regions where reciprocity is open, imbalanced, or driven.

3. Minimal Numerical Toy Model

We validate this formulation by simulating the coupled base transport equations on a 1D grid ($N=128$, $L=1.0$):

$$\partial_t \rho = \mathcal{A}(x,t) \left[D_\rho \nabla^2 \rho - \chi \nabla \cdot \left(\frac{\rho}{1 + \gamma \rho} \nabla C \right) \right] + \Xi_\rho$$

$$\partial_t C = \mathcal{A}(x,t) \left[D_C \nabla^2 C - \lambda_C (C - C_0) + \beta \rho \right] + \Xi_C$$

where γ is a chemotactic saturation parameter that prevents singular finite-time blow-ups.

3.1 Four Activation Regimes

The numerical simulations verify the self-starting properties of the reciprocity defect driver:

- Regime A — Closed Silent Equilibrium:** Starting from a homogeneous state with no external source, the reciprocity defect remains zero ($\delta_R = 0$). The activation throttle remains completely dead ($\mathcal{A} = 0.0$), and the system remains frozen and silent.
- Regime B — Artificial Floor:** Injecting a small spatial wave perturbation and enforcing a 10% artificial activation floor allows standard diffusion to flat-out and dissipate the perturbation in nominal equilibrium. This highlights the artificial, always-on nature of standard non-gated transport equations.
- Regime C — Local Reciprocity Defect:** Starting from a highly localized initial density perturbation with no external source, the system exhibits localized transient activation around the defect. Once the gradient-driven transport relaxes the fields back to a homogeneous state,

the defect decays back to zero, and the system dynamically freezes itself back into absolute silence.

4. **Regime D — Open-System Forcing:** Applying a weak, localized, and periodic external reciprocity source continuously perturbs local reciprocity balance, generating a persistent reciprocity defect ($\delta_R \neq 0$). The activation throttle remains localized ($\mathcal{A} \rightarrow 1$), and the density and coherence fields undergo spontaneous, self-starting, and persistent spatial pattern formation, maintaining stable activated fronts indefinitely.

4. Observations & Shannon Entropy

Instead of a local provisional entropy-like density which can go negative, we define a normalized global Shannon entropy to track macrostate evolution:

$$p_i = \frac{\rho_i}{\sum_j \rho_j}, \quad S_\rho = -\sum_i p_i \ln(p_i + \epsilon)$$

During open-system pattern formation (Regime D), $S_\rho(t)$ decreases dynamically as stable spatial aggregation self-assembles under open forcing, supplying a rigorous thermodynamic visualization of order emergence.

5. Connections

- [[FRC-100-001]] — Original coherence definition and FRC operator
- [[FRC-100-007]] — The Lambda-field as the fundamental agent of coherence dynamics
- [[FRC-566-001]] — Global entropy-coherence reciprocity law
- [[FRC-566-010]] — Universal Coherence Condition (UCC) flow formulation

6. References

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